

Project Communication Plan

Date	03 July 2026
Team ID	
Project Name	A Comprehensive Measure of Well-Being: Human Development Index (HDI) Predictor
Maximum Marks	1 Mark

1. Overview

Clear communication is key to stakeholders understanding the value, requirements, and findings of the Human Development Index (HDI) Prediction System. This document details the communication channels, reporting templates, and target audiences.

2. Target Audiences & Objectives

The project coordinates information across four distinct stakeholder groups:

Stakeholder Group	Communication Objective	Key Information Shared	Channel/Format
Government Officials	Gain policy approval and resource allocation	High-level index predictions, regional development rankings	Policy Briefs, Executive Presentations
Development Analysts	Facilitate detailed scenario planning	Feature coefficients, model validation metrics, inputs sensitivity	Technical Reports, Interactive Demo runs
Academic Researchers	Support comparative analysis	Datasets info, algorithm selection rationale, regression metrics	Academic Papers, GitHub Repositories
Project Team	Ensure alignment on sprints and milestones	Task progress, blockages, model tuning findings	Daily standups, Sprint boards

3. Communication Channels

3.1 Digital Repositories

- **GitHub Repository:** Primary repository containing code, test scripts, and markdown documentation.
- **Project Boards:** Used to track sprints, coordinate task status, and manage dependencies.

3.2 Visual Presentations

- **Interactive Web App:** Serving the Flask application locally to showcase live predictions.
- **Visual Dashboards:** Color-coded badges and gradient gauges to quickly communicate development levels to non-technical users.

4. Feedback Mechanism

Users can provide feedback on predictions directly through the web application's contact forms or repository issue trackers. This feedback is reviewed during sprint reviews to prioritize model enhancements and interface updates.

Communication Plan:

Effective communication is essential for a successful project demonstration. This document outlines the communication strategy used within the team and with stakeholders throughout the project lifecycle, including how updates, issues, and feedback were managed.

S.No	Communication Type	Frequency	Channel / Tool	Participants	Purpose
1	Self Project Planning	Daily	Personal Notes, GitHub	Sai Kiran Bandi, Yaswanth Yadav, Keerthi, Swapna Beldhari	Plan daily development tasks, documentation, and project milestones.
2	Progress Tracking	Weekly	Git & GitHub	Sai Kiran Bandi, Yaswanth Yadav	Track project progress, maintain version history,

				Keerthi, Swapna Beldhari	and monitor completed tasks.
3	Issue Resolution	As Needed	ChatGPT, Python Documentation, Stack Overflow	Sai Kiran Bandi, Yaswanth Yadav Keerthi, Swapna Beldhari	Resolve coding issues, debugging, Machine Learning implementation, and Flask integration.
4	Stakeholder Review	Bi-Weekly	Review Sessions	All Team Members & Mentor	Receive feedback on project development, testing, and documentation.
5	Final Demo Rehearsal	Once	PowerPoint & Local Flask Application	Sai Kiran Bandi, Yaswanth Yadav Keerthi, Swapna Beldhari	Demonstrate the complete HDI prediction system before final submission.

Communication Challenges & Resolutions:

S.No	Challenge Faced	Resolution / Action Taken
1	Selecting the most suitable Machine Learning model for HDI prediction.	Compared Linear Regression, Decision Tree Regressor, and KNN models and selected the Linear Regression model based on its superior interpretability (coefficients directly relate to UNDP dimensions) and high performance ($R^2 = 0.8834$).

2	Integrating the trained Machine Learning model with the Flask web application.	Serialized the trained model and LabelEncoder using Pickle, integrated them into the Flask backend, and verified prediction functionality through extensive local testing.
3	Ensuring reliable user input validation and prediction accuracy.	Implemented both client-side validation (HTML5 attributes) and server-side validation (Flask) to prevent invalid inputs (e.g. negative life expectancy or GNI) and improve application reliability.